Performance Analysis of a realistic
Cloud Radio Access Network (C-RAN)

Optical point-to-point system performance

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1 Introduction

In this report, we develop a fiber-link simulator for the communication between servers and a basestation. Given this context, we start by modelling a standard singlemode fiber (SSF), considering the loss and chromatic dispersion it introduces in the signals that propagate through it. Therefore, we start by defining the transfer function of the fiber, given by equation Equation 1.

$$H_{\mathrm{CD}}(\Delta\omega) = \exp\left(-\alpha L - i\left(\frac{\beta_2\Delta\omega^2}{2} + \frac{\beta_3\Delta\omega^3}{6}\right)L\right) \tag{1}$$

In this equation, α is the attenuation constant, β_2 and β_3 are the second and third order terms of the Taylor expansion of the propagation constant of the fiber around $\Delta\omega$ (Equation 2), $\Delta\omega$ is the frequency offset (Equation 3) and L is the fiber length.

$$\beta(\omega) = \beta_0 + \beta_1\Delta\omega + \frac{\beta_2}{2}\Delta\omega^2 + \frac{\beta_3}{6}\Delta\omega^3 + \ldots \tag{2}$$

$$\Delta\omega = \omega - \omega_0 = 2\pi\Delta f \tag{3}$$

And β_2 and β_3 are given by Equation 4 and Equation 5, respectively.

$$\beta_2 = -\frac{D_\lambda\lambda^2}{2\pi c} \tag{4}$$

$$\beta_3 = \frac{S\lambda^4}{4\pi^2c^2} \tag{5}$$

Now, λ is the wavelength, c is the speed of light, D is the dispersion at a given wavelength and S is the dispersion slope. The dispersion D can be calculated by Equation 6.

$$D_\lambda = D_{\lambda_0} + S(\lambda - \lambda_0) \tag{6}$$

Implementing the fiber simulation through this set of equations, we proceed to study the system performance simulating the section marked in red in the transmission chain in Figure 1. We perform tests considering external modulation and direct modulation, using different data rates, distinct fiber lengths, and varying the polarization current of the laser.

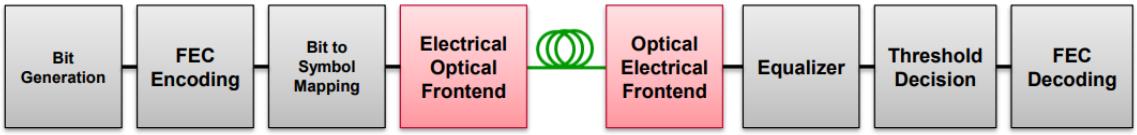


Figure 1: Complete optical transmission chain.

We then finalize our analysis by suggesting different solutions to transmit at 10 Gbit/s over 100 km with an uncorrected BER lower than 10^{-3}.

2 System Implementation

2.1 Simulation of linear propagation in a lossy and dispersive fiber

We start by implementing the equations detailed in the previous section. Considering values for a standard singlemode fiber (described in the ITU G.652 specification) in the reference wavelength $\lambda_0 = 1550$ nm , we have that $\alpha = 0.2$ dB/km, $D_{\lambda_0} = 17$ ps/nm/km and $S = 0.056$ ps/nm^2/km.

Effectively, according to our laboratory measurements, the laser emits at the wavelength $\lambda = 1554$ nm, which lets us determine D_λ, β_2 and β_3 using the equations in the introduction. This is implemented in the simulation code.

This is sufficient to simulate the optical fiber behavior. However, we seek to compare the simulation results with an analytical solution of the pulse spread $\Delta\tau$ (through Equation 7), to validate the code.

$$\Delta\tau = LD\Delta\lambda \tag{7}$$

To determine the spectral occupation $\Delta\lambda$, we must first define which pulse to emit with the laser. In this section, we consider a Gaussian pulse of full width half maximum (FWHM) T in the time domain. Then, calculating its FWHM in the frequency domain, it is found to be $\Delta f = \frac{0.44}{T_0}$. The FWHM is measured from the intensity profiles.

Using this result, the spectral occupation can be calculated as $\Delta\lambda = \frac{\Delta f \lambda^2}{c}$. Letting us determine the expected pulse spread.

Continuing with the description of the code, we implemented a function that generates a vector of samples of a Gaussian with its intensity FWHM as a chosen T_0, also calculating a sampling frequency that leads to minimal amount of aliasing. This Gaussian vector is then passed to the fiber simulation function, which performs an FFT to get its baseband frequency domain representation.

Then, it calculates the transfer function (described in Equation 1) value at every frequency bin of the resulting vector, considering that the Gaussian is centered at $\lambda = 1554$ nm. Lastly, it multiplies the frequency domain Gaussian by the transfer function, performing the IFFT and obtaining the output pulse, which can then be measured to determine the pulse spread.

Thus, we test the code first considering a Gaussian pulse of width 2 ps transmitted over a fiber of 1 km, with the same physical properties as detailed in the beginning of the section. Figure 2 shows the input and output pulses, where we can clearly notice the effects of chromatic dispersion.

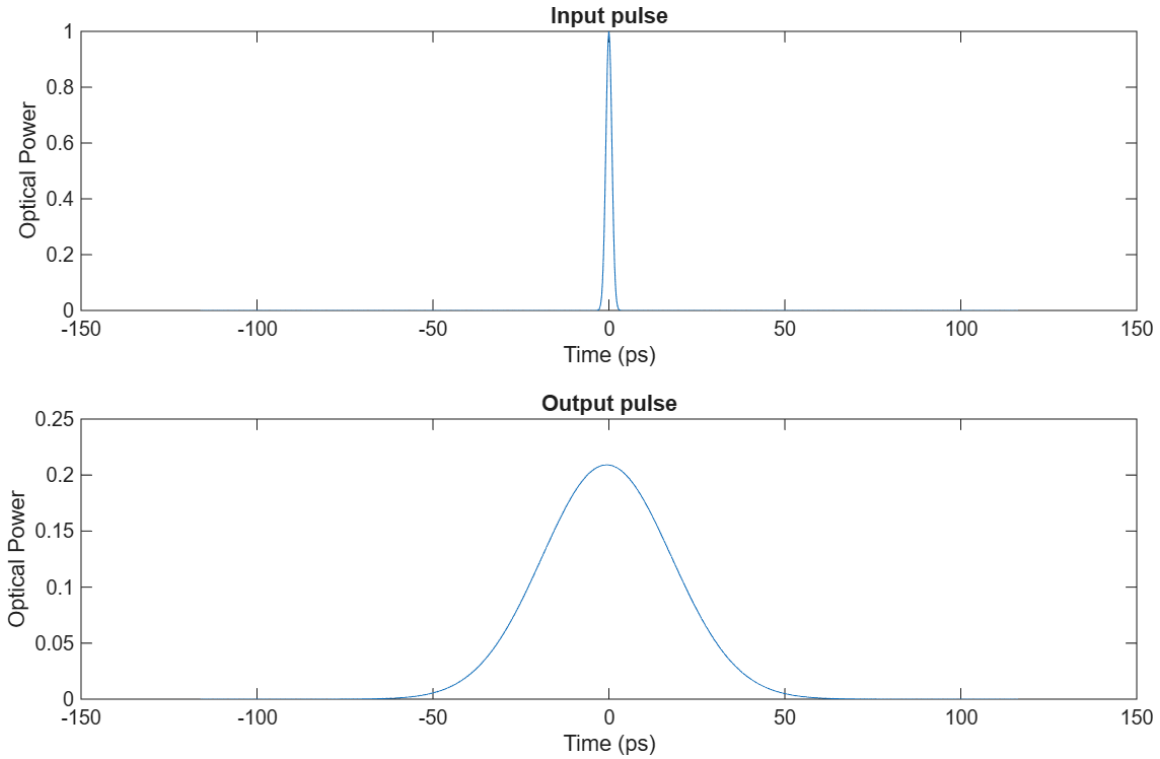


Figure 2: Gaussian pulse intensity as a function of time at the input of the fiber and at its output. Considers a pulse of width $T_0 = 2$ ps.

Comparing the pulse spreads, we get that the expected spread determined analytically is $\Delta\tau_a = 30.5$ ps, while the one measured in the simulation is $\Delta\tau_s = 30.65$ ps. The difference is less than 1%, showing that the simulated fiber performs as predicted in theory.

Testing again with a pulse width of 5 ps, the resulting pulses can be seen in Figure 3. Comparing the spreads, we expected from theory $\Delta\tau_a = 12.2$ ps and got $\Delta\tau_s = 13.2$ ps. This increase can be explained by the influence of the input pulse's own width, which is not negligible compared to the spread in this case.

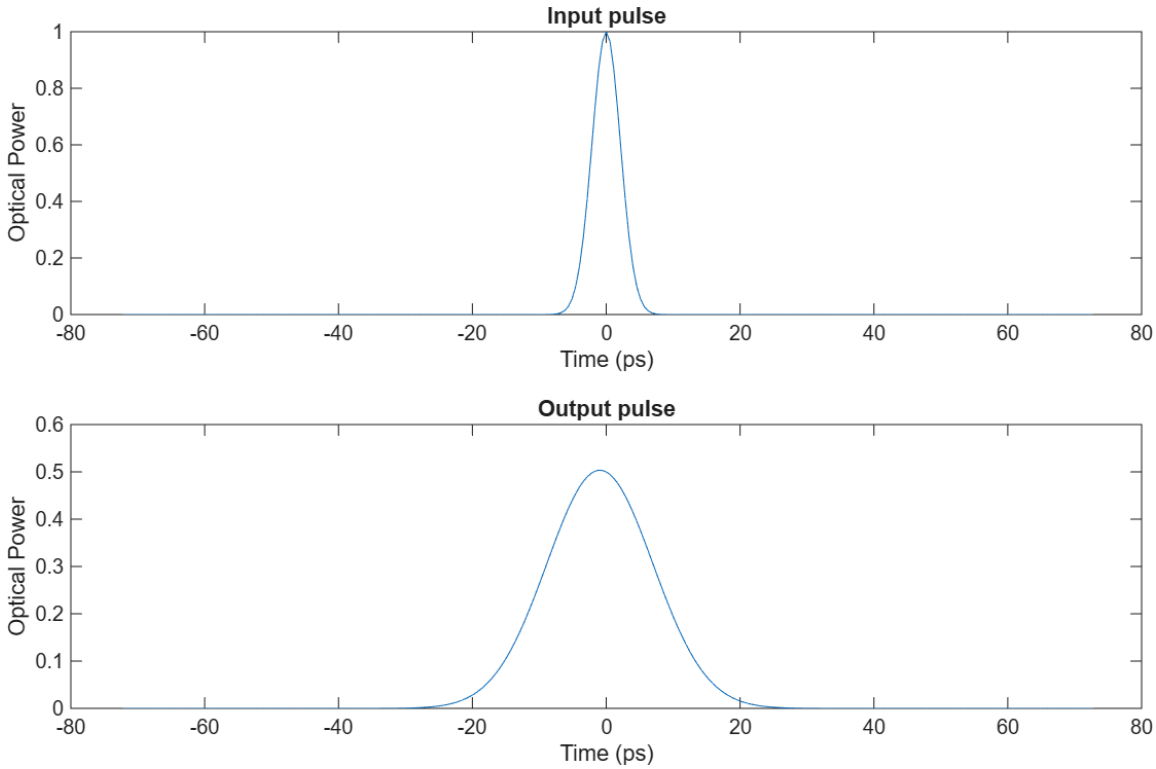


Figure 3: Gaussian pulse intensity as a function of time at the input of the fiber and at its output. Considers a pulse of width $T_0 = 5$ ps.

Therefore, we can conclude that the fiber simulation is properly implemented and proceed with the system simulation.

As a final addendum, Figure 4 shows the phase and magnitude of the transfer function simulating the optical fiber's behavior. Observe that the phase quickly increases as $\Delta\omega$ increases, while the magnitude is constant.

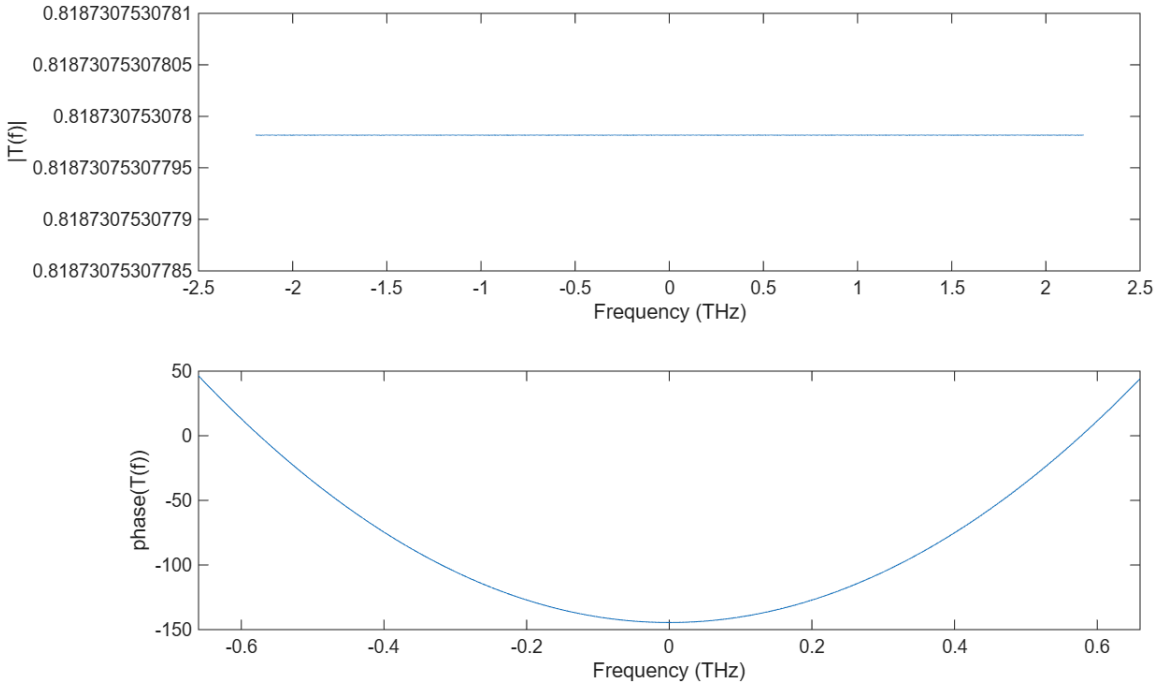


Figure 4: Transfer function of the optical fiber.

2.2 Back-to-back performance with external modulation

Now, we proceed with the testing of the back to back performance of the system, considering an externally modulated laser transmitting directly into a photodiode. In this setup, the transmission link does not employ optical fiber, such that the signal experiences neither optical attenuation nor chromatic dispersion (and consequently, no Inter-Symbol Interference). The emitter uses a Non Return to Zero (NRZ) On Off Keying (OOK) modulation scheme.

The simulations will be performed considering datarates of $R_b = 2.5$ Gb/s and $R_b = 10$ Gb/s. The aim is to determine the minimum necessary optical power P_{min} to achieve a BER of 10^{-3} and 10^{-6} in each case.

2.2.1 Analytical Considerations

Under this project's conditions, the Bit Error Rate (BER) is directly related to the noise added at the receiver. For a PIN photodiode without optical pre-amplification operating at low received optical powers, as is the case here, the dominant noise source is the thermal noise of the receiver photodiode.

Therefore, the thermal noise added by the photodiode has variance σ_{th}^2, given by Equation 8.

$$\sigma_{\text{th}}^2 = \frac{N_{\text{th}} \cdot B_e}{R_{\text{load}}} \tag{8}$$

In this equation, $N_{\text{th}} = 4k_BT$ is the thermal noise power spectral density, B_e is the electrical bandwidth of the receiver filter, and R_{load} is the load resistance. To limit the thermal noise, B_e is set equal to the bit rate R_b in each simulation. The temperature is considered $T = 290$ K and the load resistance is $R_{\text{load}} = 50\ \Omega$.

Continuing the analysis, the binary error rate can be calculated through Equation 9, where the factor Q can be determined from Equation 10. Observe that I_1 is the photocurrent and σ_1 is the noise standard deviation when a bit 1 is transmitted, while I_0 and σ_0 correspond to a bit 0.

$$\text{BER} = \frac{1}{2}\ \text{erfc}\left(\frac{Q}{\sqrt{2}}\right) \tag{9}$$

$$Q = \frac{I_1 - I_0}{\sigma_1 + \sigma_0} \tag{10}$$

Since we employ OOK modulation, we may consider $I_1 = SP_{\text{peak}}$ and $I_0 = 0$, where $S = 1$ A/W is the photodiode's sensitivity and P_{peak} is the peak optical power. Similarly, we consider that the thermal noise acts symmetrically on both bits, such that $\sigma_0 = \sigma_1 = \sigma_{\text{th}}$.

Since we search the minimum optical power necessary to achieve a certain BER, we rewrite everything as Equation 11.

$$P_{\text{peak}} = \frac{2Q}{S}\sqrt{\frac{4k_B T B_e}{R_{\text{load}}}} \tag{11}$$

Now, we proceed to analytically find the necessary powers. From Equation 9 we determine that the necessary Q factor to achieve a 10^{-3} error rate is $Q \approx 3.09$, while to achieve a 10^{-6} error rate we need $Q \approx 4.75$.

Then, solving Equation 11 for both datarates and BERs, we get Table 1.

R_b (Gbps)	BER	$P_{\text{peak}}(\mu W)$	P_{peak}(dBm)
2.5	10^{-3}	5.45	-22.6
2.5	10^{-6}	8.50	-20.7
10	10^{-3}	10.9	-19.6
10	10^{-6}	16.7	-17.8

Table 1: Minimum necessary power to achieve the desired BER at each datarate.

2.2.2 Simulation

Having determined the expected values from theory, we now implement the simulation that will calculate the BER for a series of power levels, permitting us to determine the necessary power for the desired BERs.

The simulation is conducted by generating a random vector of bits, inputting it into the `TX_optical_eml` to generate the modulated OOK optical signal, and upsampling the signal such that it better approximates a continuous signal. Then, we convolve the result with a rectangle function with the same size as the upsampling factor (guaranteeing NRZ), representing the pulses sent when the bit is 1. At last, this transmitted signal passes onto the `RX_photodetector` function, generating an electrical signal vector, which is sampled to determine the received bits.

At last, the sent and received bits are compared, and the errors measured. A loop repeats the process until the BER measure is sufficiently precise, and for each desired power level. The respective code is in the `extern_mod_main` file.

The graph of BER per Power generated by the algorithm is shown in Figure 5. Observe that the datarates of $R_b = 2.5$ Gb/s, in blue, and $R_b = 10$ Gb/s, in orange, differ in 3 dB because of the different electrical bandwidths used at the photodiode.

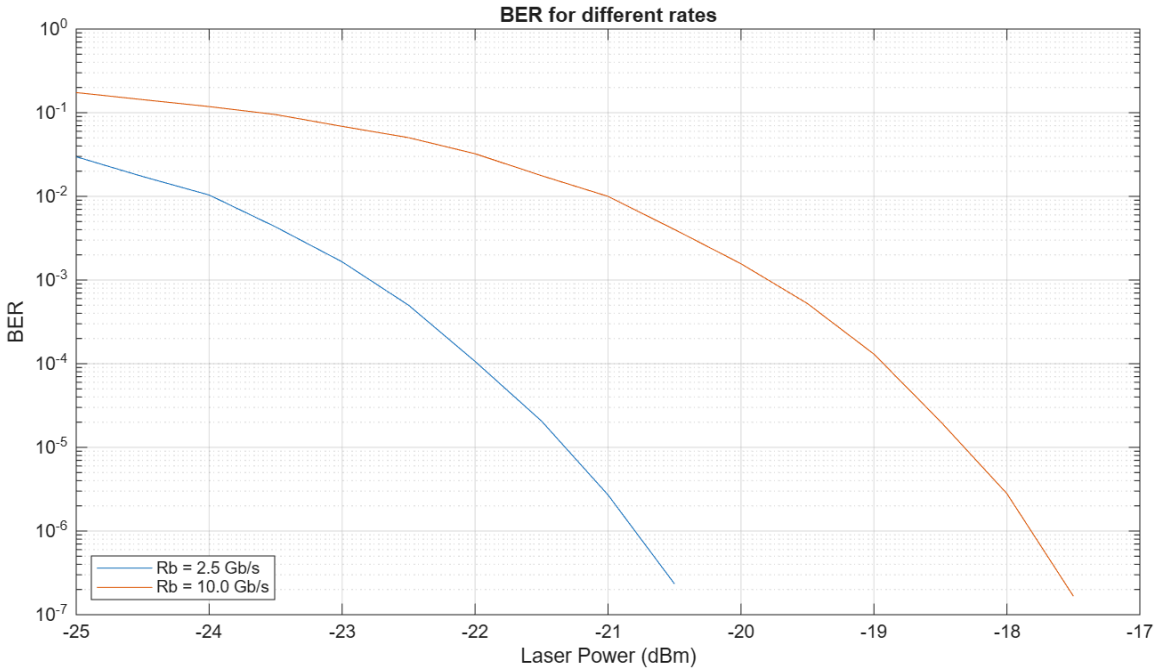


Figure 5: BER as a function of the laser power in dBm, considering datarates of $R_b = 2.5$ Gb/s, in blue, and $R_b = 10$ Gb/s, in orange.

Analyzing the simulation results, they agree almost exactly with the expected analytical values.

Thus, we may conclude that the optical system was accurately implemented in the code.

2.3 Transmission performance with external modulation

In this section, we fix a certain power level for the laser and vary the length of optical fiber between the transmitter and the receiver, seeking to evaluate the impact of chromatic dispersion on the BER. We employ the same two datarates as before (2.5 Gb/s and 10 Gb/s).

A general upper limit to the chromatic dispersion, before it significantly affects the signal, is defined as $\Delta\tau \ll \frac{T}{4}$, where T is the symbol period. Therefore, we can use Equation 7 to find a theoretical maximum length $L_{\max}$ for the fiber before chromatic dispersion leads to an increase in BER. This limit is shown in Equation 12.

$$L_{\max} = \frac{c}{2D\lambda^2\Delta f^2} \tag{12}$$

Therefore, considering that a rectangle pulse has a spectral occupation $\Delta f = \frac{2}{T}$, and substituting the known values in the above equation, we may calculate $L_{\max}$ for both datarates. Thus, for 10 Gbps we expect $L_{\max} = 9$ km, while for 2.5 Gbps we expect $L_{\max} = 146$ km, 16 times higher.

However, Optical Networks: A Practical Perspective by Ramaswami, Sivarajan and Sasaki propose an empirical limit given by $R_b^2 L < 3607$ Gbps^{2}km, considering a 2 dB

power penalty with respect to the BER in an externally modulated NRZ OOK laser. This gives $L_{\max} \approx 36$ km for 10 Gbps or $L_{\max} \approx 577$ km for 2.5 Gbps. The calculation method is effectively the same, but considering that spectral occupation is the same as the datarate and that the upper limit for $\Delta\tau$ is $\approx \frac{T}{2}$ instead of $\frac{T}{4}$ for the 2 dB penalty.

Finally, we can proceed with the simulations. The laser power is fixed at −21 dBm for the 2.5 Gbps datarate and −17 dBm for the 10 Gbps datarate. The fiber attenuation is set to 0, such that the laser power is conserved for all distances and does not influence the results.

In this context, the optical chain is simulated for fiber lengths between 0 and 100 km, calculating the BER at each. The results are shown in Figure 6.

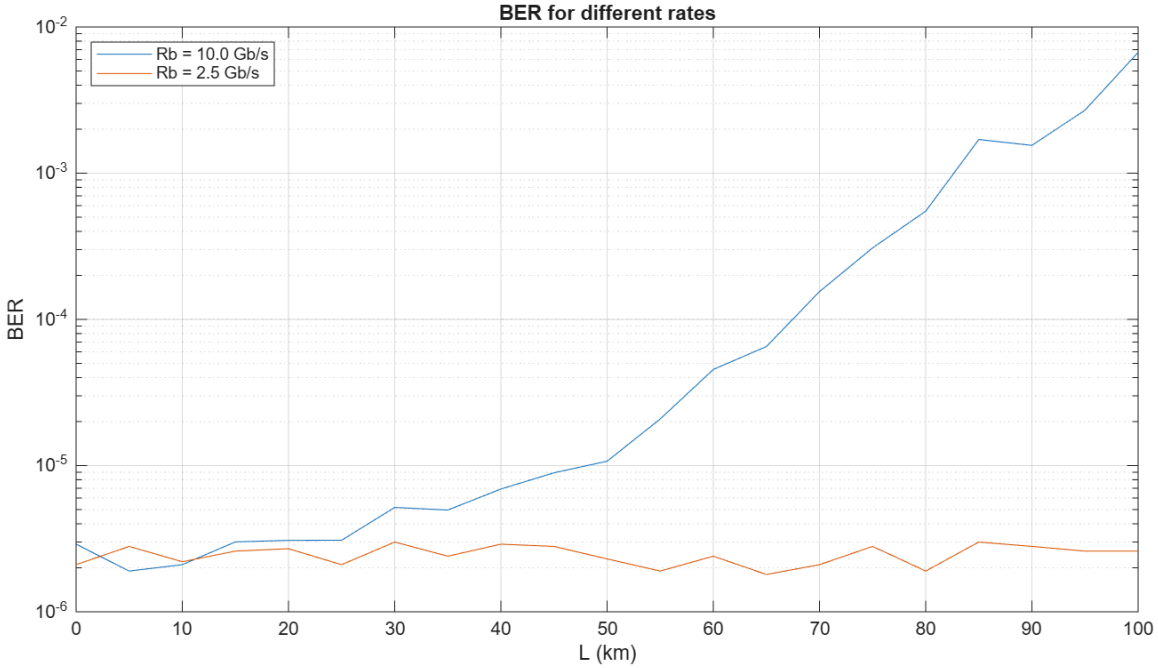


Figure 6: BER as a function of the fiber optic length in km, considering datarates of $R_b = 2.5$ Gb/s, in orange, and $R_b = 10$ Gb/s, in blue.

From the graph, we observe that the fiber length (within the 100 km bound) does not influence the 2.5 Gbps in any way, agreeing with the theoretical $L_{\max}$ we previously calculated. However, for the 10 Gbps datarate, we observe that the BER pronouncedly starts to increase at 30 km, spiking upwards a few kilometers after.

This shows us that the assumptions that led us to define $L_{\max} = 9$ km for the 10 Gbps case were too conservative, but still within the right magnitude since we truly observe a spike in BER within the observation window. This discrepancy can be attributed to, among other factors, the shaping filter we applied on the square pulse for this simulation (motivated by the terrible properties we observed when submitting the square pulse to chromatic dispersion in our tests).

Adopting the 2 dB penalty definition, $L_{\max} \approx 80$ km in our simulations. Even if distinct from the theory, our results mostly agree with the commonly adopted max range of the ITU G.652 fiber without dispersion compensation.

2.4 Performance with direct modulation

As the last simulation, we employ a direct modulation of the laser with an OOK signal of 1 Gbps, on a back to back configuration and using a 20 km fiber. The objective is to study how different bias currents I_{DC} impact on the modulation and, consequently, on the system performance.

To achieve this end,

2.5 Solutions for higher datarates and reach

3 Conclusion

Throughout this report we discussed